

Q1)

A)

Faster R-CNN: Make CNN do proposals!

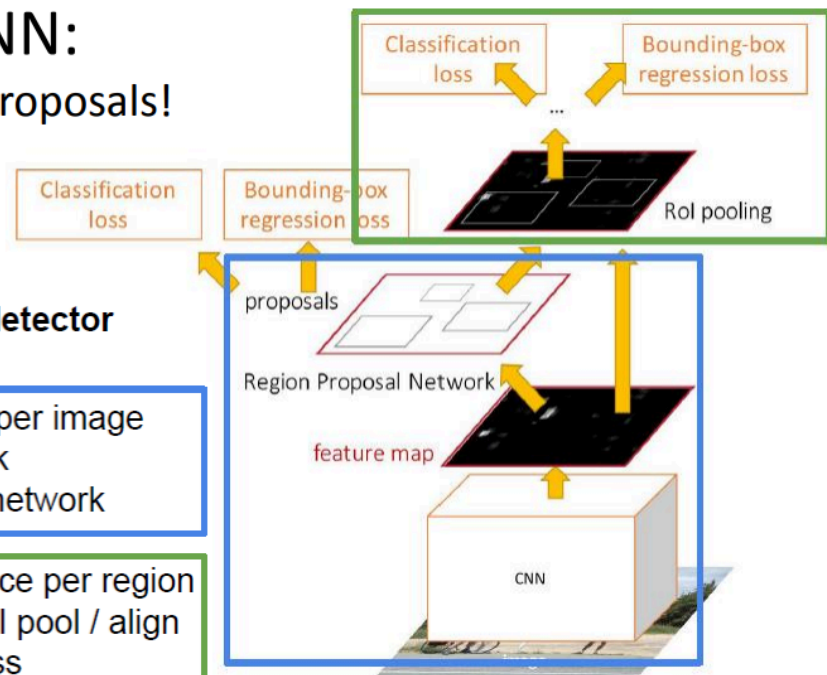
Faster R-CNN is a
Two-stage object detector

First stage: Run once per image

- Backbone network
- Region proposal network

Second stage: Run once per region

- Crop features: RoI pool / align
- Predict object class
- Prediction bbox offset



(lec 9, slide 51)

B)

HoG features would be more suitable, since SIFT features are more suitable for image matching because they describe key points in the image while the HoG describe the whole image.

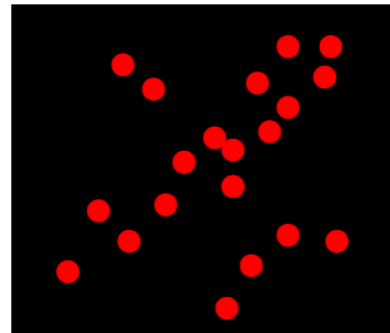
Q2)

A)

I.

RANSAC

1. **Sample** (randomly) the number of points required to fit the model.
2. **Solve** for model parameters using sample.
3. **Score** by the fraction of *inliers* within a preset threshold of the model.
4. **Repeat** 1-3 until the best model is found with high confidence.



(lec 6, slide 24)

II.

$$e = 1 - \text{inliers} = 1 - 0.65 = 0.35$$

$$P = 0.9$$

$$s = 2 \text{ (fitting a line requires min. 2 points)}$$

$$N > \frac{\log(1-p)}{\log(1-(1-e)^s)} = \frac{\log(1-0.9)}{\log(1-(1-0.35)^2)} = 4.19 \approx 5$$

B)

- I. It's the matrix computed from image derivatives and can show presence Edges or corners in the specific area it was calculated on.

Good Corners are regions where I_x^2 and I_y^2 are large

II.

$$R = \det(M) - \alpha \text{trace}(M)^2 = ad - bc - \alpha(a + d)$$

$$R = 403 \times 381 - 385 \times 385 - 0.04(403 + 381)^2 = -19268.24$$

مش علينا Q3)

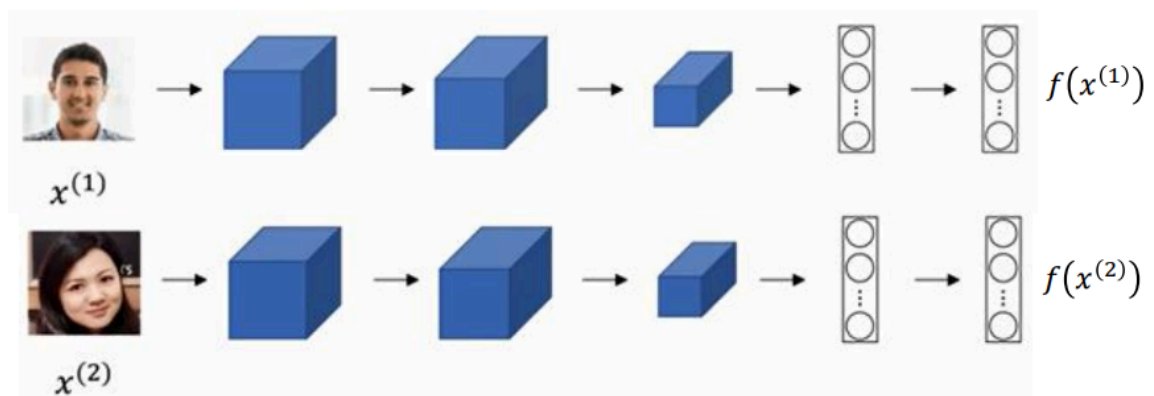
Q4)

مش علينا A)

B)

Architecture:

Siamese Network



$$d(x^{(1)}, x^{(2)}) = \|f(x^{(1)}) - f(x^{(2)})\|_2^2$$

If $x^{(i)}, x^{(j)}$ are the same person, $\|f(x^{(i)}) - f(x^{(j)})\|_2^2$ is small.

If $x^{(i)}, x^{(j)}$ are different persons, $\|f(x^{(i)}) - f(x^{(j)})\|_2^2$ is large.

(Lec 8, Slide 48)

The triplet loss aims to Minimize distance between anchor and positive images and maximize distance between anchor and negative images

$$L(A, P, N) = \max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0)$$